

1 Verilog Implementation

```
1 module phase1(  
2     input  clk,  
3     input  rst,  
4     input  code_in,  
5     output reg phase1_done,  
6     output reg phase1_fail  
7 );  
8  
9 parameter safe = 3'b000;  
10 parameter s1  = 3'b001;  
11 parameter s2  = 3'b010;  
12 parameter s3  = 3'b011;  
13 parameter s4  = 3'b100;  
14 parameter done = 3'b101;  
15 parameter fail = 3'b110;  
16  
17 reg alarm;  
18 reg [2:0] current_state, next_state;  
19  
20 always @(posedge clk) begin  
21     if(rst)  
22         current_state = safe;  
23     else  
24         current_state = next_state;  
25 end  
26  
27 always @(*) begin  
28     phase1_done = 0;  
29     phase1_fail = 0;  
30     alarm = 0;  
31  
32     case(current_state)  
33  
34         safe:  
35             next_state = s1;  
36  
37         s1:  
38             if(code_in)  
39                 next_state = s2;  
40             else  
41                 next_state = fail;  
42  
43         s2:  
44             if(!code_in)  
45                 next_state = s3;  
46             else  
47                 next_state = fail;  
48  
49         s3:
```

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50         if(code_in)
51             next_state = s4;
52         else
53             next_state = fail;
54
55     s4:
56         if(code_in)
57             next_state = done;
58         else
59             next_state = fail;
60
61     done:
62         phase1_done = 1;
63
64     fail: begin
65         phase1_fail = 1;
66         alarm = 1;
67     end
68
69     default:
70         next_state = safe;
71
72     endcase
73 end
74
75 endmodule

```